



Computer Infrastructure Capstone –

CPSC 49300

Syllabus

Spring Semester 2024

I. Instructor Information

Instructor's name: Sean Sullivan

Office hours: By appointment and virtual

How to make appointments outside of office hours: via E-Mail or Phone

Lewis email address: ssullivan7@lewisu.edu

II. Course Information

Course Name, Number, & Section: Comp Infra Capstone – CPSC 49300-002/003/0

Course Credit Hours: 3

Course description: In this course students will participate, as part of a team, in the design, implementation and testing of a medium-to-large networked computer system. Additionally, this course will cover topics in professional ethics, intellectual properties, privacy and professional communication.

Supplementary Course Description: Students will conduct all necessary research to make informative decisions on the survey, design and implementation process. Students conclude their project by producing a final well-written and structured paper. The final paper will document the entire process of project completion, including problem definition, requirements, specifications, design, implementation, and tutorial. Students will present their papers to their advisor, faculty and peers.

PREREQUISITE: Senior Standing and CPSC-28200 or CPSC-48500.

Course meeting times, days, and location: See Blackboard announcements but classes will be recorded, if you cannot attend. You will have group sessions for the group you are assigned to for your capstone project.

Student Learning Outcomes:

- a) Formulate a networking or security related problem to solve, that provides ample room for investigation and knowledge gathering, and is narrow enough to allow for in-depth coverage of the issues.
- b) Conduct a thorough review of the literature on the chosen area that provides sufficient coverage of previous scholarly work.
- c) Develop skills of collecting project requirements and specifications.

- d) Design, develop, implement, test and document IT solutions for networking and security problems, balancing business concerns, technical issues and security.
- e) Plan, schedule, manage, and use best practices of network design and implementation.
- f) Identify and address ethical and legal issues that arise in network design and implementation.
- g) Develop the skills of working within a team as a member of an IT project team.
- h) Apply learning outcomes of previous computer science courses in general and in networking and security courses in particular to an IT project
- i) Identify service and civic responsibility as personal and professional priorities.
- j) Appropriately cite related past work to establish claims and make justified points.
- k) Produce a high quality and well-structured document to reflect the outcomes learned above.
- l) Present key issues and findings clearly and in an engaging way to an audience of peers.

Program student learning outcomes:

- 1. Explain how computers are organized and implemented in hardware and software
- 2. Work in a team to design and implement complex data processing systems collaboratively
- 3. Describe how computer systems can communicate securely with each other and form networks.
- 4. Document project requirements and specifications, ethical issues, policies, related theoretical and technical background, references, models, plans, designs, and implementation process.

Baccalaureate Characteristics/Graduate Student Learning Outcomes:

- 1. Essential Skills
- 2. Major Approaches to Knowledge
- 3. Moral and Ethical Decision-Making
- 4. Critical Thinking
- 5. Lifelong Learning
- 6. Production of a high-quality paper that includes all learning outcomes.

III. Instructional Methods and Activities

Modality of Instruction: Online, meetings by arrangements

Additional Activities:

Students may work with not for profit organizations partners to complete IT tasks for these organizations

IV. University Mission Statement

Lewis University, guided by its Catholic and Lasallian heritage, provides to a diverse student population programs for a liberal and professional education grounded in the interaction of knowledge and fidelity in the search for truth. Lewis promotes the development of the complete person through the pursuit of wisdom and justice.

Fundamental to its Mission is a spirit of association, which develops community in all teaching, learning and service.

How this course connects to the University Mission:

This course embraces the Mission of the University by fostering an environment in which each student is respected as an individual within a community of learners, where students work individually and in groups to develop knowledge and wisdom, and to represent this development by a high-quality white paper. In the spirit of the vision of Lewis University, the goals and objectives of this course seek to prepare students to be successful, life-long learners who are intellectually engaged, ethically grounded, socially responsible, and globally aware.

V. Required Course Materials

Textbook(s):

No textbook is required for this course.

Hardware and software requirements:

Laptop with internet access is required

VI. Grading Criteria and Course Policies

Grade Components

The course grade will be determined by the following:

Discussion (10 – 10 pts each)	100 points	
Project Milestones	400 points	
Individual Report	100 points	
Final Group Report	200 points	
Presentation	200 points	
Total Points	1000 points	100%

Grade	Percentage
A	100% to 93%
A-	92% to 90%
B+	89% to 87%
B	86% to 83%
B-	82% to 80%
C+	79% to 77%
C	76% to 73%
C-	72% to 70%
D+	69% to 67%
D	66% to 63%
D-	62% to 60%
F	59% or less

Grading Criteria

Grade A, A- Requirements

1. Demonstrates accurate and sophisticated understanding of readings and issues. Does more than repeat text or lectures. Draws out additional important implications.
2. Shows a critical stance toward opinions expressed in class or readings. Expresses their own views articulately and defends them well.
3. Shows originality of thought in pressing the critical stance, in drawing out additional implications from readings and class discussions, and in finding personal meaning in the readings and issues discussed.
4. Expressed ideas clearly. Papers contain very few grammatical or stylistic weaknesses.
5. Demonstrates a leadership role in team assignments and attends all teamwork sessions.

Grade B+, B and B- Requirements

1. Shows all elements of "A" work but will less accomplishment.
2. There is still accurate understanding of readings/issues with a demonstrated ability to do more than repeat text.
3. Critical stance with some effort, not always successful to defend that stance.
4. Some attempt to find personal meaning, with at least hints of originality and creativity of thought.
5. Very clear expression of thoughts and ideas.
6. Contributes more than their required portions in team assignments and participates in all team work sessions.

Grade C+, C and C- Requirements

1. Overall, demonstrates an accurate grasp of the readings and issues, but with some inaccuracy and without deeper sophistication and the ability to draw out additional implications.
2. Some attempt to take a critical stance, but with little effort or success in defending that stance.
3. Some attempt to find personal meaning.
4. Sufficient clarity of expression to communicate ideas but stylistic or grammatical weakness creates some difficulties.
5. Performs their portion of team assignments thoroughly and accurately / participates in most team work sessions.

Grade D+, D and D- Requirements

1. A genuine effort to understand has been made, with some demonstrated understanding of reading and issues but with serious deficiencies.
2. Generally lacking in critical stance or in a defense of that stance.
3. The attempt to find personal meaning is either lacking or greatly hindered by difficulties in understanding.
4. Despite some errors, performs most of their portion of team assignments thoroughly and accurately / participates in some team work sessions.

Grade F Requirements

1. Simply an unsatisfactory effort in key respects, especially in understanding and clarity of expression.

Changes to Course Assignments or Grades: Instructor will notify students of any changes to course assignment or grades in a timely fashion.

Course Policies:

Project and Assessment Process

The course project has two components, a theory and background component, and a service-learning component. Projects will be completed by groups.

In the theory and background component each group is required to use at least ten (10) resources to cover essential theoretical material in networking, data communication, and security. Each group is expected to write number of pages equivalent to 7 times the number of group members specifically for this component, illustrating and describing essential topics in networking and security. If a group has three students, then 21 pages are expected for this component. In the service-learning component, each group will work on an infrastructure project that will assist a community organization in implementing their mission.

See ***Final Group Report Requirements*** for project requirements and structure.

Option 1: Groups will meet with the professor as the client to assist with a not-for-profit tutoring center that will be described.

Option 2: Groups are encouraged to help identifying potential clients through their connections. Priority goes to a not-for-profit organization, however if a not-for-profit organization is not available, and a student knows of a corporation that needs help with their infrastructure, then this option will be considered for a potential project. Examples of the types of projects that you might work on include developing a network system for a school, church, charity organization, or any not-for-profit organization, where this network provides wired and wireless access for local and remote communication, and for Internet access. If working with a client, you will need to visit the site of the project, meet and communicate with the client on regular basis throughout the semester. It is expected that the major part of the project to be completed at the client's side. If a client is not available through school or through students, then a student or group of students not to exceed three students may select a project of their own.

Each group is expected to write number of pages equivalent to 12 times the number of group members for both components (a theory and background component, and a service-learning component). If a group has three students, then at least a total of 36 pages is expected for both components excluding title, abstract, table of contents and references.

Students are expected to provide the following:

- a- **Update/Reflection:** Students will submit a written update/reflection each week describing their completed activities for the week.
- b- **Milestones:** Students will maintain a working outline of their paper throughout the course. The outline will contain major entries for each section of the report, and subtopics in the outline will correspond to subsections or paragraphs within each report section. The outline will be reflected and checked through each milestone's submission. These milestones must be recognized in high-quality and well-structured documents providing details of all completed activities. Students will use supplementary material and links as guidance for the contents of the milestone documents. Material and links used should be properly cited.
- c- **Final Group Report (White Paper):** Towards the end of the semester each group will demonstrate the functionality of the developed system and submit a white paper documentation of the completed project. This document shall include both components of the project, the theory and background component, and the service-learning component showing project requirements and specifications, ethical issues, policies, related theoretical and technical background, models, plans, designs, and implementation process. Details and structure of the final report are provided in a special section below.
- d- **Individual Reports:** Each student is expected to submit a separate individual report summarizing her/his contribution to project milestones, and to the final project and final report. A considerable weight will be allocated to the contents, organization, spelling and grammar, depth and focus, cohesiveness, and citations of the submitted reports.
- e- **Project Presentation:** each group with the presence and participation of all group members will present their final document to the instructor and students. Other instructors and department Chair will be invited to attend the presentation. Each group will have 30 to 40 minutes for the presentation depending on the number of students in each group, approximately 10 minutes for each student.

A discussion forum will be created for students to post questions, answer questions, and provide input and feedback.

Your performance will be assessed against the completion of tasks as scheduled, including weekly reports, project milestones' accomplishments and reports, final project completion, final report completion, individual reports, demonstrations of completed tasks, the interaction with your group, the client and the instructor throughout the process, and the writing skills of submitted reports and papers. The following course outline shows how a total of 1000 points are distributed over the different tasks throughout the semester.

Individual Report

Each student is expected to provide an individual report of three to five pages stating the following:

- 1) Description of the project, this could be your own summary of the group report
- 2) The contribution of each team member to the actual project and to the final group report, you need to include your contribution and the contributions of your teammates
- 3) Your own reflection on the project including:
 - a. The project itself,
 - b. Dealing with clients,
 - c. Providing community-based learning,
 - d. Working with a team
- 4) Significant weight will be assigned for structure, grammar and spelling.

Final Group Report Requirements

Since your capstone project is a group project, **I expect one report for the entire group.** This report shall be made from the periodic milestone reports submitted during the semester.

However, it has to be written as one final report and not as a group of sub-reports.

Group project shall include the following:

- 1) A title for the project to be determined by the group
- 2) An abstract about the project
- 3) Table of Contents
- 4) **Theory and Background**; this component must include the following:

- a. Introduction to networking including, LAN, MAN, WAN, and Internet
 - b. LAN technologies, WAN technologies, and Wireless technologies
 - c. OSI reference model and TCP/IP model including protocols, and devices associated with each model.
 - d. Security attacks, and security mechanisms for LAN and WAN environments.
 - e. VPN technology
 1. Site-to-site
 2. Remote access
 - f. Provide a model for a company with three different branches in three different cities, where each branch has 10 to 20 users, and these branches are sharing database, documents, and applications. All these branches must have Internet access. Provide a solution, showing all required devices at all three sites, describe devices models and features, what technologies are used at each site, what technologies are used to connect the different sites, how Internet access is provided. Show what security measures are used to secure all resources including data, applications, communications and devices. You may use the cloud for some services, but you can't depend completely on the cloud.
 - g. The above model should be described and explained in detail writing and in pictographic.
 - h. Cite all your references in APA 7th edition, and list all references in the references section. For this part you are expected to use at least 10 different resources NOT provided by the instructor.
 - i. This part is expected to be fully completed and included in the second milestone.
- 5) An introduction providing more details about the project and the organization
 - 6) Include literature review
 - i. <https://www.lewisu.edu/writingcenter/pdf/final-lit-review-resource-pdf.pdf>
 - 7) Approach: provide a description for how the project was approached, include the different types of information that was collected including information about:
 - a. The organization
 - b. Users including administration, staff, and others
 - c. Physical location including buildings, offices, classrooms, and closets

- d. Existing equipment
 - e. Budget
 - f. Security concerns
 - g. Other information you encountered and collected
- 8) Provide pictures for before and after
 - 9) Physical and logical topologies for the entire network. Topologies shall be detailed and descriptive. This part will have an important weight for the final grade.
 - 10) Your report must have a section about ethical issues you encountered, and you addressed through the project development
 - 11) Describe the design, implementation, testing, security, and documentation process.
 - 12) Describe how you balanced business concerns and needs, with budget constraints.
 - 13) Describe how did you apply learning outcomes of previous computer science courses in general and in networking and security courses in particular to this project.
 - 14) Your report should be in the form sections or chapters, concluded with a conclusion section.
 - 15) Setup the layout of the report showing the different section and sub-sections.
 - 16) Support your report with pictures, tables and figures.
 - 17) Your report must be well organized and free of any spelling and grammatical errors.
 - 18) Group report is expected to be between 25 to 35 pages in total.
 - 19) Individual report is expected to be 3 to 5 pages in total.
 - 20) **References:** you must identify your references using APA style. List only the references you cite in the paper; do not list references you do not cite in the body of the paper. *Liberally cite. There is no such thing as excessive citation.* Opinions, statistics, and claims that are not yours or that are not general knowledge in the strictest sense all must be cited.

Format: The paper must be written with the following formatting:

- 12-point font
- Double-spaced
- 1" margins
- First line of paragraphs indented
- Consistent chapter and subsection headings

- Table captions centered above the table, formatted like "Table 1. Attacks by month"
- Figure captions centered below the figure, formatted like "Figure1. Attacks by month"
- Tables and figures referenced in the text before appearing in the paper.
- Avoid personal pronouns (ideally, there should be none)
- APA formatting (see Lewis APA Style Reference guide and Purdue OWL website if unfamiliar with APA)([apa-7-guide-final-2.12-pdf.pdf](#) [APA Style Introduction - Purdue OWL](#))
- **Submit both WORD document and PDF document**

Grading Criteria

All sections of the final paper will be graded based on the following rubric:

Criteria	Description	Score
Grammar, spelling, punctuation, syntax, and mechanics:	<i>Your paper is free of grammar and punctuation errors, and all words are spelled correctly. Your writing is clear and concise. Starting sentences with the subject, use of active verbs, minimize adjectives and adverbs, use short forms of words and phrases.</i>	30%
Organization	<i>Your paper is organized into the sections identified in this syllabus (introduction, review of literature, project analysis and design, configurations, results, conclusion, citations). Table, figure, and page numbers align. Title page, abstract, and table of contents appropriately structure the paper. Avoid extraneous ideas and focus on one idea at the time.</i>	10%
References	<i>Your paper sufficiently surveys the existing literature regarding your project. Citations are consistently formatted and properly referenced in the body of the paper.</i>	20%
Content	<i>Your paper is technically and conceptually sound. You have made appropriate use of the skills and concepts you learned in your classes. Your paper describes the actual completed project and isn't padded with extraneous material that has little to do with the project. You avoid making generalizations and carefully support every claim you make, whether by citing other experts or reporting the results of your implementations.</i>	40%

Your final presentation will be graded based on the following rubric:

Criteria	Description	Score
Presentation Style:	<i>You speak clearly, at a good pace, and in an engaging way. You avoid reading the slides and instead use them only to keep the</i>	20%

Criteria	Description	Score
	<i>audience focused on the ideas you are presenting. Your slides aren't wordy, are attractive, and keep the audience focused.</i>	
Organization	<i>You organized the topics logically and in a way that facilitates understanding. You present the high points of your paper with enough detail to foster understanding and prompt further questions.</i>	20%
Contents and products	<i>Quality and completion of project, and contents for both components</i>	40%
Logistics	<i>Your presentation was completed on-time, allowing at least five minutes for questions. Your slides are legible on the presentation PC. Technical issues do not cause you to stop your presentation</i>	10%
Questions	<i>You are able to answer questions from the audience well and don't try to fake an answer you do not know.</i>	10%

Policies and Procedures

Academic Honesty: Scholastic integrity lies at the heart of Lewis University. Plagiarism, collusion and other forms of cheating or scholastic dishonesty are incompatible with the principles of the University. Students engaging in such activities are subject to loss of credit and expulsion from the University.

Attendance: This class doesn't not have a scheduled meeting time; however weekly updates are mandatory. It is your responsibility to manage your project and keep it on schedule.

Sanctified Zone: This learning space is an extension of Lewis University's Sanctified Zone, a place where people are committed to working to end racism, bias and prejudice by valuing diversity in a safe and nurturing environment. This active promotion of diversity and the opposition to all forms of prejudice and bias are a powerful and healing expression of our desire to be Signum Fidei, "Signs of Faith," in accordance with the Lewis Mission Statement. Learn more about the [Sanctified Zone here](#).

Late Assignment Policy: It is expected that all assignments will be submitted on time unless prior arrangements (before the due date) are made. An extension is not automatically granted. Assignments turned in late will have a 5% per day reduction in grade. Assignments submitted later than one week after the deadline will receive no credit.

Drop and Withdrawal Deadlines: course may be dropped with a full refund in the first week and a partial refund in the second week. You may withdraw from this course through the end of the 10th week. For specific details and more information, see: [Office of the Bursar](#)

[University Student Complaint Policy](#)

The University Student Complaint Policy can be found at lewisu.edu/studentcomplaints

[University Grade Appeal Policy](#)

The University Grade Appeal Policy can be found at lewisu.edu/studentcomplaints

[University Copyright and Intellectual Property Guidelines](#)

The University Copyright and Intellectual Property Guidelines can be found at lewisu.edu/academics/library/pdf/Copyright-Intellectual-Property-Guidelines

Note: Potential Modifications

The instructor reserves the right to modify, change, or waive any part of the syllabus or the evaluation criteria for this course. The instructor will provide prior notification via email or Bb announcement of any modifications.

By Lewis policy and practice, the Lewis email system will be the preferred method of communication to students. Students should normally expect a response within 24 to 48 hours.

VII. Information for Students

Requests for Reasonable Accommodations

Lewis University is committed to providing equal access and opportunity for participation in all programs, services and activities. If you are a student with a disability who would like to request a reasonable accommodation, please speak with the Learning Access Coordinator at the Center for Academic Success and Enrichment (CASE). Please make an appointment by calling 815-836-5593 or emailing learningaccess@lewisu.edu. For more information about academic support services, visit the website at: www.lewisu.edu/CASE. Since accommodations require early planning and are not provided retroactively, it is recommended that you make your request prior to or during the first week of class. It is not necessary to disclose the nature of your disability to your instructor.

Sanctified Zone

Guided by its Catholic and Lasallian heritage, Lewis University is firmly committed to fostering a campus atmosphere that is permeated by its Mission values of Fidelity, Wisdom, Knowledge, Justice, and Association. Accordingly, we have declared the University campus to be a Sanctified Zone, a place and a people *United in Diversity*. The active promotion of diversity and the opposition to all forms of prejudice and bias are a powerful and healing expression of our desire to be Signs of Faith (Signum Fidei) to each other. To learn more about the Sanctified Zone, please visit: <http://www.lewisu.edu/sanctifiedzone>.

Academic Integrity

Scholastic integrity lies at the heart of Lewis University. Plagiarism, collusion and other forms of cheating or scholastic dishonesty are incompatible with the principles of the

University. Students engaging in such activities are subject to loss of credit and expulsion from the University. Cases involving academic dishonesty are initially considered and determined at the instructor level. If the student is not satisfied with the instructor's explanation, the student may appeal at the department/program level. Appeal of the department /program decision must be made to the Dean of the college/school. The Dean reviews the appeal and makes the final decision in all cases except those in which suspension or expulsion is recommended, and in these cases the Provost makes the final decision.

Center for Health & Counseling Services

To support student success, all Lewis students are eligible for free health and mental health services on the Romeoville campus. This includes commuters and those living on campus, part-time and full-time students, graduate and undergraduate students, and those taking Lewis classes at other locations. For more information, visit the Center for Health & Counseling website at www.lewisu.edu/studentservices/health or call (815)836-5455 for more information.

VIII. Practices and Policies during the Coronavirus Pandemic

Responsiveness to Change

Understanding that the COVID-19 pandemic could influence the course of this semester, Lewis University will be guided by our Lasallian mission and the well-being of our community of students, faculty, and staff in respond and adapting to any sudden changes or circumstances. Based on the guidance of the State of Illinois and the Centers for Disease Control, it may be necessary to require adherence to new to health and safety protocols and/or make changes to the planned modality this course.

Flexibility, Accommodations, and Student Absences

The University community is committed to academic standards while maintaining flexibility and empathy.

- In accordance with the Flyer's Promise, students must report positive COVID-19 tests and exposures to Lewis University. Visit the Lewis COVID homepage for reporting instructions.
- Absences relating to the Coronavirus crisis will require documentation to be recognized as excused.
- Students experiencing disruptions in their lives related to the Coronavirus that impact class attendance and participation should contact their instructor to make arrangements for completing missed work.

Students who require academic accommodations due to disability caused by COVID-19, or to limit risk of exposure to Coronavirus, can engage in an interactive process with the Learning Access Coordinator to explore avenues for accommodations. Students can contact the Academic Services office at 815-836-5593 or learningaccess@lewisu.edu to request an appointment.

Except in the case of unforeseen circumstances that require a university-wide shift in instructional modalities, courses will only be offered in the modality listed in the course schedule. It is not possible to accommodate remote attendance in on-campus classes unless the course is listed as multi-option in the schedule.

Face Coverings

The Lewis community is a mask friendly environment and recognizes the benefits of wearing face coverings by encouraging and supporting all who choose to wear one.

- Some settings on campus may require face coverings. When this is the case, clear signage will be posted.
- Based on the guidance of the State of Illinois and the Centers for Disease Control, it may be necessary for Lewis to require face coverings. Lewis will notify the campus community if policy changes.

IX. Course Schedule

Schedule Changes: the instructor will notify student within 48 hours of schedule changes.

Day	Topics/Assignments/Quizzes/Exams
Week 1 1/21 – 1/26	<u>Introduction</u> • Discussion Board: Introductions (10 points) • Discussion Board: Syllabus Discussion (10 points) • Identifying groups, groups' members, clients and projects • Students are encouraged to find clients from a not-for-profit organization • Process understanding • Review material for the Theory and Background component and start writing
Week 2 1/27 – 2/2	<u>Site visit and evaluations</u> • Site visit and meet with client • Read supplementary material and links provided by your instructor as guidance • Evaluate existing infrastructure • Take inventory of existing equipment • Identify users and understand their needs • Learn about the budget allocated for the project • Understand and protect business privacy and confidentiality • Ensure all project goals and requirements are legal and legitimate • Review material for the Theory and Background component and continue writing

Day	Topics/Assignments/Quizzes/Exams
	Discussion Board: Written Report of Site Evaluation and progress on the theory and background component (10 points)

Week 3 2/3 – 2/9	<u>Analysis</u> - Read supplementary material and links provided by your instructor as guidance - Intensive meetings with client and users - Understand their needs and expectations - Understand the physical layout of the site - Identify locations, rooms, closets where equipment will be installed or relocated - Identify needed equipment - Assign responsibilities to team members - A team leader should be finalized in coordination with the instructor - Review material for the Theory and Background component and continue writing - Discussion Board: Documentation of Analysis, and progress on the theory and background component (10 points)
Week 4 2/10 – 2/16	<u>Design and proposal</u> - Read supplementary material and links provided by your instructor as guidance - Design the physical topology - Provide an estimate of the cost - Provide a description of the new topology and the functionalities - Make recommendations for ISP for Internet access - Include the above items in a proposal - Present the proposal to client (Milestone-1, 150 points) Submit a well-written and structured document for the entire team covering the first 4 weeks of progress, and a proposal detailing the goals of the project. The theory and background component must include the following: ➤ Introduction to networking including, LAN, MAN, WAN, and Internet ➤ LAN technologies, WAN technologies, and Wireless technologies ➤ OSI reference model and TCP/IP model including protocols, and devices associated with each model ➤ Security attacks, and security mechanisms for LAN and WAN environments ➤ VPN technology ➤ Cite all your references, and list all references in the references section. For this component you are expected to use at least 5 different resources NOT provided by the instructor - Cite any material you use in your process. The report must include all team members and a summary of their contributions.

Day	Topics/Assignments/Quizzes/Exams
Week 5 2/17 – 2/23 Week 6 2/24 – 3/2 Week 7 3/3 – 3/9	<u>Physical topology implementation</u> • Read supplementary material and links provided by your instructor as guidance • Identify wirings runs through walls, above the ceilings or under floorings • Physical topology implementations shall start without causing business hours interruption, or with minimal and tolerated interruption if necessary • Additional equipment and wiring shall be purchased if needed • Disposal or recycling of old equipment if any shall be according to health, environmental, safety, and privacy guidelines • Installing new machines and wiring • Relocating old machines and wiring • Ensure the physical security of intermediary and critical devices such as routers, switches, servers and firewalls • Install WAP as needed • Generate all necessary documentation for users and technical administrators • Week 5 – Discussion Board: Physical Topology 1 (10 points) • Week 6 – Discussion Board: Physical Topology 2 (10 points) • Complete the theory and background component as follow: ○ Provide a model for a company with three different branches in three different cities, where each branch has 10 to 20 users, and these branches are sharing databases, documents, and applications. All these branches must have Internet access. Provide a solution, showing all required devices at all three sites, describe devices models and features, what technologies are used at each site, what technologies are used to connect the different sites, how Internet access is provided. Show what security measures are used to secure all resources including data, applications, communications and devices. You may use the cloud for some services, but you can't depend completely on the cloud. ○ The above model should be described and explained in detail writing and in pictographic. ○ Cite all your references, and list all references in the references section. For this component you are expected to use at least 10 different resources in total NOT provided by the instructor. • (Milestone-2, 150 points) At the end of Week 7 submit a project mile stone report reflecting the first seven weeks. Cite any material you use in your process.

Day	Topics/Assignments/Quizzes/Exams
Week 8 3/10 – 3/16 BREAK 3/17 – 3/23 Week 9 3/24 – 3/30 Week 10 3/31 – 4/6	<u>Configurations</u> • Read supplementary material and links provided by your instructor as guidance • Develop the logical topology showing the different subnets • Configure all machines including routers, servers, workstations, and any end device with IP addresses, default gateway and DNS • Add all necessary configurations to all machines • Configure firewalls • Configure security measures on all machines • Add all users accounts • Configure access rights to all users • Generate all necessary documentation for users and technical administrators • Week 8 – Discussion Board: Configuration 1 (10 points) • Week 9 – Discussion Board: Configuration 2 (10 points) • (Milestone-3, 50 points) Use the previous project milestone report and update it with tasks completed in weeks 8, 9, & 10, submit at the end of Week 10. Cite any material you use in your process.
Week 11 4/7 – 4/13	<u>Testing</u> • Read supplementary material and links provided by your instructor as guidance • Test functionalities, performance and reliability of the infrastructure • Test wireless coverage to ensure it meets client's expectations • Monitor traffic to ensure load balancing and appropriate configuration • Make updates as necessary • List any tools you used for testing • Discussion Board: Submit a report detailing the results of Testing/Troubleshooting (10 points)
Week 12 4/14 – 4/20	<u>Training</u> • Read supplementary material and links provided by your instructor as guidance • Train clients on how to use the system and the documentations • Train the administrator and technical team if any on how to use the system, and introduce them to the system documentations for future reference • Train users on how to use the system • Seek users' feedback on the system • Discussion Board: Provide detailed instructions to serve as training for users (10 points)

Day	Topics/Assignments/Quizzes/Exams
Week 13 4/21 – 4/27	<u>Organization policies</u> • Read supplementary material and links provided by your instructor as guidance • Work with the client on setting privacy and security policies • Document those policies • Educate client and staff on those policies • Seek users' feedback on the system • (Milestone-4, 50 points) Use the previous project milestone report and update it with tasks completed in weeks 11, 12 & 13. Cite any material you use in your process.
Week 14 4/28 – 5/4 Week 15 5/5 – 5/11 Week 16 5/12 – 5/17 *Semester ends Sat, 5/17	<u>Demonstration, Presentation, Individual Report and Final Report</u> • (200 points) Team members shall use all previous documents and reports to finalize the final report. Final report structure, format and contents shall follow the specific guidelines provided by the instructor. • (200 points) Team members will present their completed project to the instructor, each group will be allocated number of minutes equivalent to 10 times the number of group members, where each member will have to present for 10 minutes. • (100 points) Each student will provide an individual report following the requirements stated below. *All due by Sunday, May 11, 2025

NOTE: The Class Schedule provides a general guideline for planning and preparation purposes only. Actual dates may vary at the instructor's discretion reflecting subject difficulty, length of discussions, addition of supplemental materials, etc.